

Mathematics

Year 7

Student Manual



Prime Books • Mathematics Year 7 • Student Book

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Printed for Prime School • Year 7 • Mathematics

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INTRODUCTION

How this book works

Welcome to Prime Books Mathematics for Year 7. This book is written for you, the pupil, to read, argue with and write in. It is not a book to be afraid of. Every idea is introduced with a reason before it becomes a rule, and every rule is followed by the chance to use it until it feels obvious.

Mathematics at this level is about making the familiar strange and the strange familiar. You already know how to add, subtract, multiply and divide. In this first unit you will stretch those four operations across the whole number line, into the territory of the negatives, and then use them to unlock older and deeper patterns: multiples, factors, quick tests for divisibility, and the roots that undo powers.

What makes this book different

- **Every concept has a reason.** You will rarely meet a bare rule. You will meet a situation that forces the rule into being.
- **Every example is worked in full.** Nothing is left as "an exercise for the reader" in the middle of a new idea.
- **You are asked to think like a mathematician.** Several times in each unit you are invited to investigate, conjecture and justify, not just calculate.
- **Everything is real.** Names, places and stories are drawn from life in Portugal, and every fact can be checked in the Sources & references at the back.

A note to teachers and parents

This book follows the spirit and sequence of an international lower-secondary mathematics programme, interpreted afresh for Prime School. Each unit opens with diagnostic questions to surface prior knowledge, builds concepts through worked examples and structured practice, and closes with self-assessment and a summative check. The pedagogy is deliberately investigation-led: pupils are asked to notice patterns before being handed formulas, because mathematics that is discovered is mathematics that is remembered.

HOW TO USE THIS BOOK

Reading the page

Each unit follows the same rhythm, so you always know where you are. Here is what each kind of panel is for.

PANEL	WHAT IT DOES
Get started	A few quick questions to wake up what you already know. No new ideas here.
Scenario	A real situation that makes the unit's maths worth learning.
Learning outcomes	The promises this unit makes to you. Tick them off as you master them.
Warm up	A short puzzle to get your mathematical blood moving.
Do you remember?	A gentle reminder of earlier work the new idea depends on.
Learn	The main teaching. Read slowly; this is where new ideas are born.
Worked example	A problem solved in full, with the reasoning shown beside it.
Practise	Questions for you. Fluency first, then reasoning, then a stretch.
Did you know?	A true and surprising fact connected to the topic.
Think like a mathematician	An open investigation. Conjecture, test, justify.
Go further	For when you want more: a harder idea or a wider view.
Challenge yourself	The stretch. Attempt these; they are meant to be hard.
Final project	A bigger task that pulls the whole unit together.
Check your progress	A mixed set of questions across the whole unit.
Evaluation	Be honest with yourself: green, amber or red on each outcome.
What can you do?	A checklist to confirm what you can now do.
Keywords	The mathematical words this unit has taught you.

How to work

Read with a pencil in your hand. When the book asks a question, stop and try it before you read the answer. Mathematics is not a spectator sport: you learn it by doing it, getting it wrong, and seeing why.

GETTING SET UP

Before Unit 1

You do not need much for this unit: a pencil, a ruler, a rubber, and a squared exercise book. A calculator is useful for checking, but most of this unit is about doing without one, so keep it in your bag until you are asked.

Your mathematical toolkit for this unit

- A **number line** you can sketch quickly, with zero in the middle.
- Times tables up to 12×12 , fluent enough that you do not have to think.
- The four operations on positive whole numbers, secure.
- Curiosity. This is the most important tool.

A quick self-check

Try these now. If any feel hard, tell your teacher before you begin: the unit builds on them.

- 1 Write the numbers -3 , 5 , 0 , -7 , 2 in order, smallest first.
- 2 List the first four multiples of 7 .
- 3 List all the factors of 20 .
- 4 Work out 13×4 and $96 \div 8$ without a calculator.

The answers, and the ideas behind them, are exactly what Unit 1 will sharpen. When you can do these confidently, you are ready.

UNIT 1

1

Integers

The whole number line, in both directions

© LEARNING OUTCOMES

- ✓ Add and subtract with positive and negative integers.
- ✓ Multiply and divide with positive and negative integers.
- ✓ Find the lowest common multiple of two or more numbers.
- ✓ Find the highest common factor and use it to simplify fractions.
- ✓ Use tests for divisibility to spot factors quickly.
- ✓ Connect square and cube numbers to their roots.

◆ GET STARTED

Try these before you begin. They wake up the ideas this unit builds on.

- 1 Write these numbers in order, smallest first: -4 , 7 , 0 , -9 , 3 .
- 2 Write down the first five multiples of 8 .
- 3 List all the factors of 30 .
- 4 Work out 15^2 , then work out $15^2 - 14^2$. What do you notice?

◆ SCENARIO

Every January, the Serra da Estrela, the highest mountains in mainland Portugal, can drop well below freezing at night, while the valleys below stay several degrees above. A weather station records a night-time low of -6°C at the summit and an afternoon high of 9°C in the village below. How many degrees separate the two readings? To answer that, you have to count across zero, which means leaving the comfortable world of positive numbers behind. That is the work of this unit.



Figure 1.2 Above and below zero, on the same scale.

Counting is one of the oldest things humans do, and for most of history the numbers we counted with were all we had: one sheep, two sheep, three sheep. Zero took thousands of years to be accepted as a number in its own right, and negative numbers took even longer. Yet the moment you want to talk about temperature below freezing, a bank balance that is overdrawn, or a floor below ground, you need numbers on both sides of zero.

The **positive whole numbers**, the **negative whole numbers** and **zero** together are called the **integers**. On a number line they stretch away in both directions, and the ellipsis at each end means they never stop.

... -4 -3 -2 -1 0 1 2 3 4 ...

The integers on a number line. Zero sits in the middle; positives to the right, negatives to the left.



Figure 1.1 The number line as a road in both directions.

Adding and subtracting integers

Learn

Think of the number line as a road. Adding a positive number moves you to the right; subtracting a positive number moves you to the left. The same is true for negatives, with one twist that surprises everyone at first: **subtracting a negative number moves you to the right**.

Why? Because subtraction is the **inverse operation** of addition. To subtract a number, you add its **inverse**. The inverse of 3 is -3 , and the inverse of -8 is 8. So the calculation $2 - (-4)$ becomes $2 + 4$, which is 6.

■ WORKED EXAMPLE 1.1

Work out:

a. $-7 + 12$ b. $5 + (-9)$ c. $-4 - (-8)$

a. $-7 + 12 = 5$ start at -7 , move 12 right
 b. $5 + (-9) = -4$ start at 5, move 9 left
 c. $-4 - (-8) = -4 + 8 = 4$ subtracting a negative adds its inverse

Why: In part c the two negatives do not cancel by magic: to subtract -8 we add its inverse, $+8$.

You can use rounding to estimate before you calculate, which is a good habit: it tells you whether your final answer is even plausible.

■ WORKED EXAMPLE 1.2

Estimate $198 + 347$ by rounding each number to the nearest ten.

$198 \rightarrow 200$ $347 \rightarrow 350$
 estimate: $200 + 350 = 550$

Why: The exact answer is 545, which sits close to our estimate of 550, so we can be confident.

• **PRACTISE 1.1**

- 1 Work out:
a. $-3 + (-6)$ b. $2 - 9$ c. $-6 + 14$ d. $-5 - (-11)$
- 2 Find the missing number: $-8 + \square = -3$.
- 3 I think of an integer, add -7 , and get 2. What number did I think of?
- 4 Write down two integers whose sum is -10 and whose difference is 2.
- 5 a. Work out $(-5) + 14 - (-3)$.
a. Explain why the order you add integers in does not change the answer.
- 6 In an addition pyramid, each block is the sum of the two below it. If the bottom row is 3, -5 , 8, find the value of the top block.
- 7 Estimate, then calculate exactly: $396 - 187 + 58$.

∴ **THINK LIKE A MATHEMATICIAN**

On a 3×3 addition table, the same three numbers head the rows and the columns. Choose the row-and-column headers -2 , 5, -7 and fill in the nine sums.

- What do you notice about the diagonal from top-left to bottom-right?
- Is the table symmetrical about that diagonal? Will it always be, whatever three headers you choose? Justify your answer.

■ **KEYWORDS**

- **integer**: a positive or negative whole number, or zero.
- **inverse**: the number that undoes another: the inverse of 4 is -4 .
- **inverse operation**: an operation that reverses another; subtraction is the inverse operation of addition.
- **number line**: a line on which numbers are placed in order.

Multiplying and dividing integers

Learn

Multiplication began as repeated addition: 3×4 means $3 + 3 + 3 + 3$. That picture still works when one factor is negative. $(-3) \times 4$ means $(-3) + (-3) + (-3) + (-3)$, which is -12 . So a negative times a positive is negative.

Division is the inverse of multiplication, so the same sign logic flows through: because $3 \times 4 = 12$, we know $12 \div 4 = 3$; and because $(-3) \times 4 = -12$, we know $-12 \div 4 = -3$. Whenever exactly one of the two numbers is negative, the **product** or quotient is negative. When both are negative, the result is positive, because the two negatives undo each other.

■ WORKED EXAMPLE 1.3

Work out:

a. $7 \times (-4)$ b. $(-9) \times (-5)$ c. $(-42) \div 6$

a. $7 \times (-4) = -28$ positive $\times$ negative = negative
 b. $(-9) \times (-5) = 45$ negative $\times$ negative = positive
 c. $(-42) \div 6 = -7$ negative $\div$ positive = negative

Why: Multiply or divide the absolute values first, then fix the sign: same signs give +, different signs give –.

When brackets appear, work inside them first.

■ WORKED EXAMPLE 1.4

Work out $5 \times (2 + (-7))$.

inside brackets: $2 + (-7) = -5$
 then: $5 \times (-5) = -25$

Why: Brackets first, always. The multiplication waits until the bracket is a single number.

• PRACTISE 1.2

- 1 Work out:
a. -6×8 b. $(-63) \div (-7)$ c. $9 \times (-5)$ d. $56 \div (-8)$
- 2 Find the missing number: $\square \times (-6) = 24$.
- 3 Find the missing number: $(-36) \div \square = 4$.
- 4 Write down all the pairs of integers whose product is -12 .
- 5 Work out $8 \times (3 - 9)$ in two different ways and check they agree.
- 6 Estimate, then calculate exactly: $(-19.7) \times 5.1$.

∴ THINK LIKE A MATHEMATICIAN

Gonalo claims that because $(-2) \times (-3) = 6$ and $2 \times 3 = 6$, "two negatives always give the same answer as two positives." Is he right that the *answers* are equal here? Does that mean the two calculations *mean* the same thing? Investigate with three more examples and explain the difference between equal answers and equal meanings.

✦ DID YOU KNOW?

The word "product" for the result of a multiplication is over four hundred years old in English mathematics. We say "12 is the product of 3 and 4" in exactly the way a merchant in 1600 would have.

■ KEYWORDS

- **product:** the result of a multiplication.
- **quotient:** the result of a division.

UNIT 1 • TOPIC 1.3

Lowest common multiples

Learn

The **multiples** of a number are what you get by multiplying it by 1, 2, 3 and so on. The multiples of 6 are 6, 12, 18, 24, 30, ... When you list the multiples of two numbers, any number that appears in both lists is a **common multiple**. The smallest of these is the **lowest common multiple**, or LCM.

■ WORKED EXAMPLE 1.5

Find the lowest common multiple of 8 and 12.

multiples of 8: 8, 16, 24, 32, 40, ...
multiples of 12: 12, 24, 36, 48, ...
common multiples: 24, 48, ...
lowest common multiple: 24

Why: A multiple of 12 must be divisible by 12; scanning the list of 8s for the first one that 12 divides finds 24 quickly.

You can find the LCM of three numbers the same way: list, compare, take the smallest common value.

• PRACTISE 1.3

- 1 List the first five multiples of:
a. 7 b. 11 c. 15
- 2 Find the lowest common multiple of:
a. 6 and 10 b. 5 and 7 c. 8 and 14
- 3 Find the lowest common multiple of 4, 6 and 9.
- 4 Write down all the common multiples of 4 and 6 that are less than 50. What do you notice about them?
- 5 Two buses leave Cascais bus station together. One returns every 18 minutes, the other every 24 minutes. After how many minutes do they next return together?

∴ THINK LIKE A MATHEMATICIAN

For the pair 4 and 6, their product is 24 and their LCM is 12. For the pair 5 and 7, their product is 35 and their LCM is 35. Investigate several more pairs. When is the LCM equal to the product of the two numbers, and when is it smaller? Can you explain why?

■ KEYWORDS

- **multiple:** the result of multiplying a number by a positive integer.
- **common multiple:** a number that is a multiple of two or more numbers.
- **lowest common multiple (LCM):** the smallest common multiple of two or more numbers.
- **digit:** any of the symbols 0 to 9 used to write numbers.

UNIT 1 • TOPIC 1.4

Highest common factors

Learn

A **factor** of a number divides it exactly. The factors of 36 are 1, 2, 3, 4, 6, 9, 12, 18 and 36, and they come in pairs that multiply to 36: 1×36 , 2×18 , 3×12 , 4×9 , 6×6 . A **common factor** of two numbers is a factor of both, and the largest of these is the **highest common factor**, or HCF.

The HCF has a beautiful use: it simplifies fractions as far as they can go. To reduce a fraction to its simplest form, divide top and bottom by their HCF.

■ WORKED EXAMPLE 1.6

- Find the highest common factor of 24 and 60.
- Hence simplify the fraction $24/60$.

factors of 24: 1, 2, 3, 4, 6, 8, 12, 24
factors of 60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60
common factors: 1, 2, 3, 4, 6, 12
highest common factor: 12
 $24 \div 12 = 2$ and $60 \div 12 = 5$, so $24/60 = 2/5$

Why: Dividing top and bottom by the HCF in one step lands you at the simplest form immediately, with no further cancelling possible.

• **PRACTISE 1.4**

- 1 List all the factors of:
a. 28 b. 45 c. 54
- 2 Find the common factors, and then the highest common factor, of:
a. 18 and 27 b. 16 and 40
- 3 Find the highest common factor of:
a. 36 and 54 b. 42 and 70 c. 30 and 75
- 4 Simplify each fraction as far as possible:
a. $18/27$ b. $16/40$ c. $36/54$
- 5 Inês has 36 pastéis de nata and 54 broas de mel. She wants to fill identical boxes, each holding only one type of pastry, with none left over. What is the greatest number of pastries she can put in each box?

∴ **THINK LIKE A MATHEMATICIAN**

Take any two numbers, find their HCF and their LCM, and multiply the HCF by the LCM. For 12 and 18: HCF is 6, LCM is 36, and $6 \times 36 = 216$. Now compute 12×18 . Try it with three more pairs. What do you discover? Does it always work? This is one of the neatest shortcuts in number work: once you see it, you can find an LCM from an HCF, or an HCF from an LCM, instantly.

■ **KEYWORDS**

- **factor**: a number that divides another exactly.
- **common factor**: a number that is a factor of two or more numbers.
- **highest common factor (HCF)**: the largest common factor of two or more numbers.

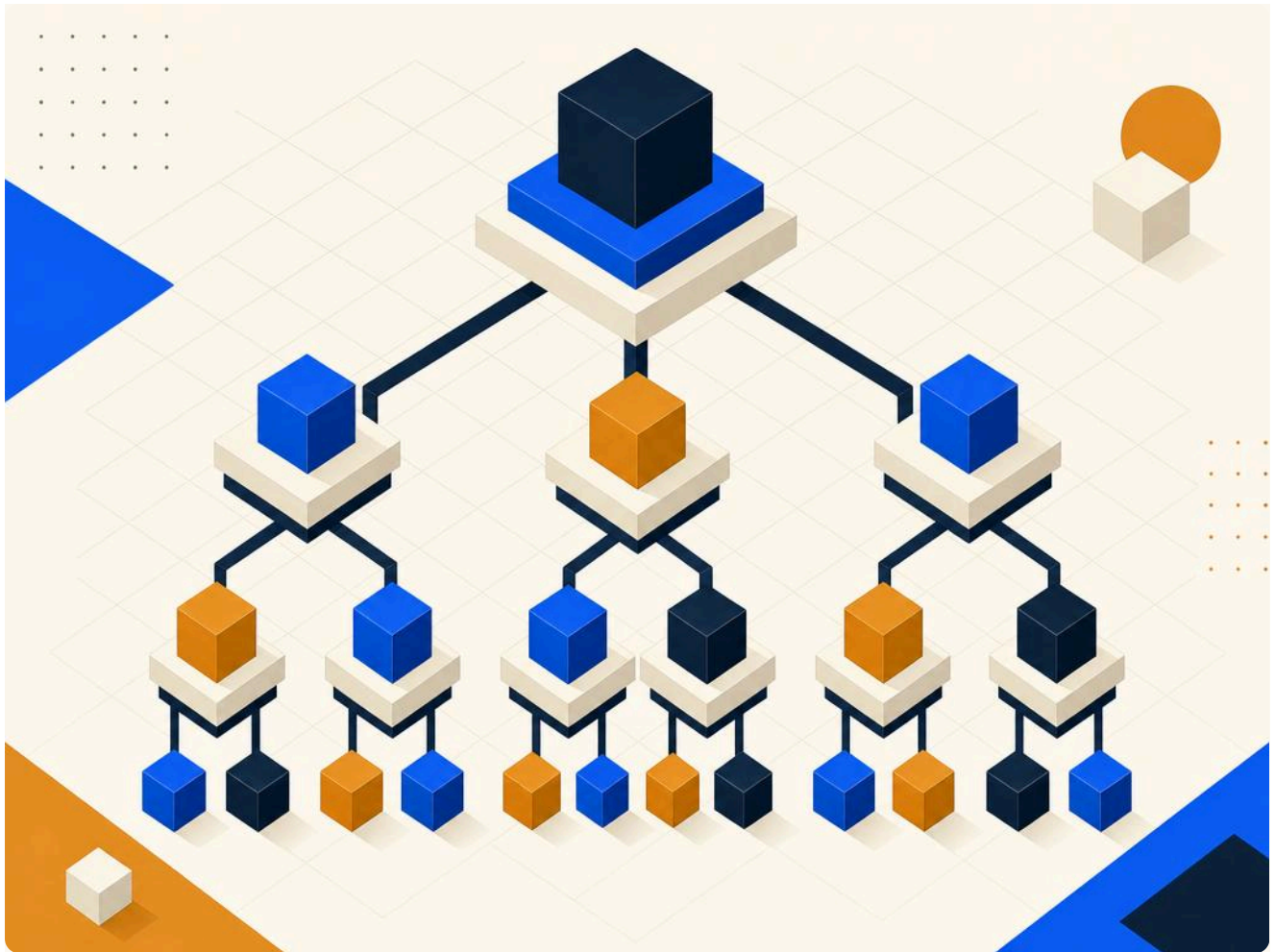


Figure 1.3 Factors branch like the limbs of a tree.

UNIT 1 • TOPIC 1.5

Tests for divisibility

Learn

A number is **divisible** by another if it divides exactly, with no remainder. For big numbers you need not divide to find out: there are quick **tests for divisibility** that use the digits themselves.

DIVISOR	TEST	QUICK CHECK ON 74 628
2	last digit even	ends in 8, so yes
3	digit sum divisible by 3	$7+4+6+2+8 = 27$, and $27 \div 3 = 9$, so yes
4	last two digits divisible by 4	$28 \div 4 = 7$, so yes
5	last digit 0 or 5	ends in 8, so no
6	passes the tests for 2 and 3	passes both, so yes
8	last three digits divisible by 8	$628 \div 8 = 78.5$, so no
9	digit sum divisible by 9	$27 \div 9 = 3$, so yes
10	last digit 0	ends in 8, so no
11	alternating digit sum is 0 or a multiple of 11	$(7+6+8) - (4+2) = 15$, so no

■ WORKED EXAMPLE 1.7

The number $\square 7258$ has a missing first digit.

- Find the digit that makes the number divisible by 3.
- Find the digit that makes it divisible by 11.

a. digit sum so far: $7+2+5+8 = 22$
 next multiple of 3 above 22 is 24, so the digit is 2 (also 5 or 8)
 b. alternating sum: $(\square+2+8) - (7+5) = \square - 2$
 for this to be 0, the digit is 2

Why: For 11, we need the two alternating sums to balance exactly, which pins the missing digit to a single value.

• **PRACTISE 1.5**

- 1 Test 91 818 for divisibility by 2, 3, 5 and 9.
- 2 Which of these numbers are divisible by 6: 5124, 7 335, 40 206?
- 3 Find the missing digit so that $5\square42$ is divisible by 9.
- 4 Use the digits 3, 4, 5 and 6 once each to make a four-digit number divisible by 11. How many different answers can you find? (One example is 3465.)
- 5 The number 44 444 is made only of the digit 4. Explain why any number made only of the digit 4 is divisible by 2 and by 4, but never by 5.
- 6 A number is divisible by both 2 and 5. What single number must it also be divisible by? Explain.

∴ **THINK LIKE A MATHEMATICIAN**

Matilde says: "If a number is divisible by 2 and by 4, it must be divisible by 8." Beatriz says: "If a number is divisible by 2 and by 5, it must be divisible by 10." Test both statements with several numbers. One of them is always true and the other is not. Which is which, and what makes the difference?

✦ **DID YOU KNOW?**

The divisibility test for 9 has a twin called "casting out nines", used for centuries by bookkeepers to catch arithmetic slips. Adding the digits of a number throws away all its nines, so if your working and your answer leave different remainders when you cast out nines, something has gone wrong.

■ **KEYWORDS**

- **divisible:** able to be divided exactly, with no remainder.
- **test for divisibility:** a quick rule that tells you whether one number divides another without doing the division.

Square roots and cube roots

Learn

When you multiply a number by itself you get a **square number**: $5 \times 5 = 25$, so 25 is a square number, and we write $5^2 = 25$. The little 2 is an **index**, and it means "multiply this number by itself". Going the other way, the number that was squared to give 25 is the **square root** of 25, written $\sqrt{25} = 5$. Squaring and taking a square root are inverse operations.

Cubes work the same way with one more multiplication: $4 \times 4 \times 4 = 64$, so 64 is a **cube number**, written $4^3 = 64$. The **cube root** undoes it: $\sqrt[3]{64} = 4$.

■ WORKED EXAMPLE 1.8

Work out:

a. $\sqrt{144}$ b. $\sqrt{169} - \sqrt{64}$ c. $\sqrt[3]{216}$

a. $12 \times 12 = 144$, so $\sqrt{144} = 12$
 b. $\sqrt{169} = 13$ and $\sqrt{64} = 8$, so $\sqrt{169} - \sqrt{64} = 13 - 8 = 5$
 c. $6 \times 6 \times 6 = 216$, so $\sqrt[3]{216} = 6$

Why: Every root is found by asking "what number, raised to this power, gives this result?"

Most numbers are not perfect squares, but you can still trap their roots between two whole numbers.

■ WORKED EXAMPLE 1.9

Between which two consecutive whole numbers does $\sqrt{50}$ lie?

$7^2 = 49$ and $8^2 = 64$
 $49 < 50 < 64$, so $7 < \sqrt{50} < 8$

Why: Because 50 sits between two consecutive squares, its root sits between the corresponding consecutive roots.

• **PRACTISE 1.6**

- 1 Write down the value of:
a. 9^2 b. 11^2 c. 6^3
- 2 Work out:
a. $\sqrt{196}$ b. $\sqrt{81} + \sqrt{25}$ c. $\sqrt[3]{1000}$
- 3 Between which two consecutive whole numbers does $\sqrt{90}$ lie?
- 4 I square a number and get 289. What was the number?
- 5 Work out $\sqrt{(81 + 144)}$. Is it the same as $\sqrt{81} + \sqrt{144}$? Explain.
- 6 A square number has an odd number of factors. Check this is true for 36 by listing its factors, then explain why it happens for every square number.

∴ **THINK LIKE A MATHEMATICIAN**

Look at the pattern of the first few cube numbers added together. $1^3 = 1$, and $1 = 1^2$. $1^3 + 2^3 = 9$, and $9 = 3^2$. $1^3 + 2^3 + 3^3 = 36$, and $36 = 6^2$. Work out $1^3 + 2^3 + 3^3 + 4^3$ and check whether the pattern continues. What is the connection between the numbers 1, 3, 6, 10 on the right-hand side? Predict the sum of the first five cubes and test your prediction.

■ **KEYWORDS**

- **square number**: a number made by multiplying an integer by itself.
- **cube number**: a number made by multiplying an integer by itself twice.
- **index**: the small raised number showing a power, as in 5^2 .
- **square root**: the number that squares to give the value under the root.
- **cube root**: the number that cubes to give the value under the root.
- **consecutive**: following one after another in order.

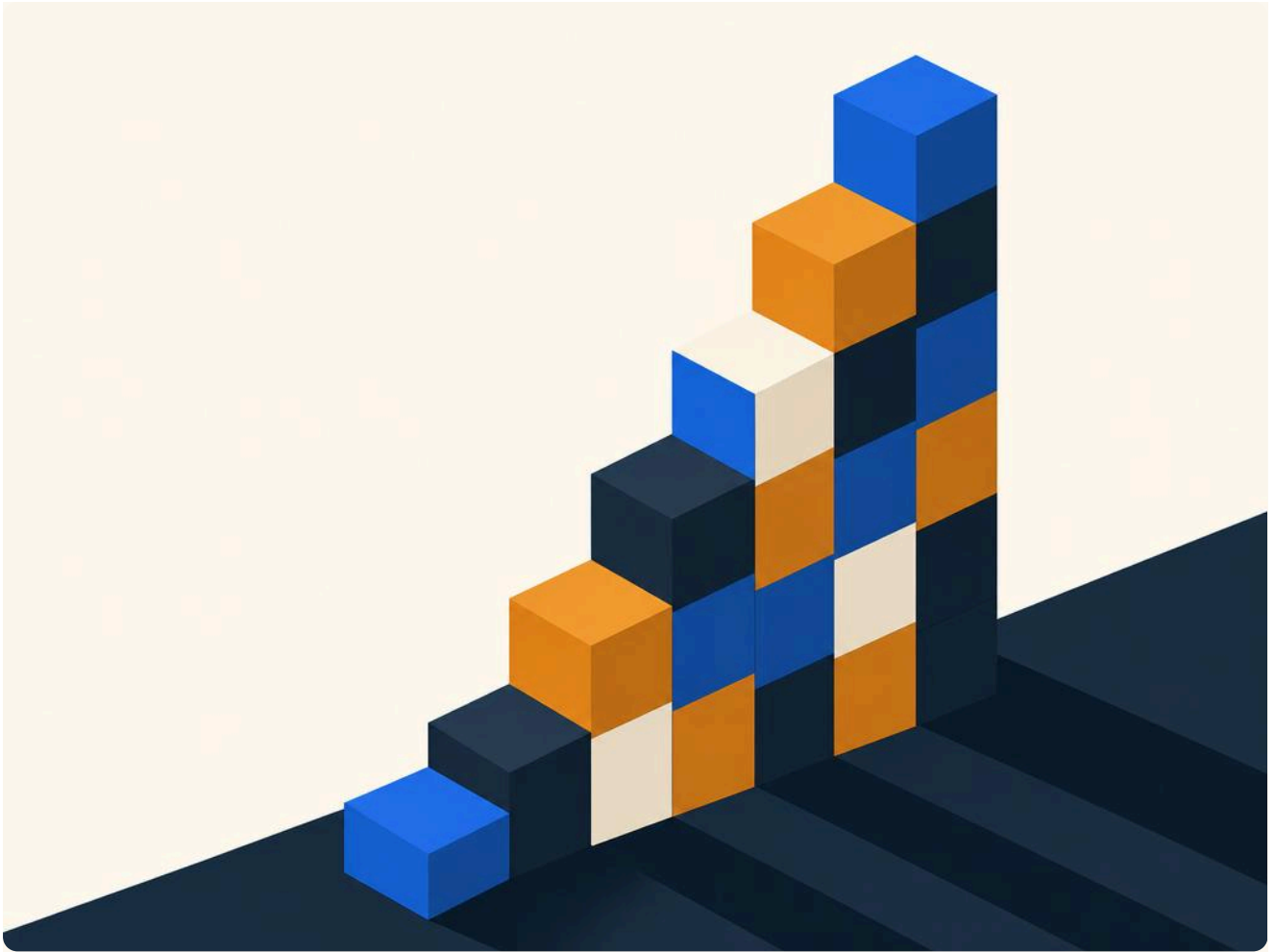


Figure 1.4 Squares and cubes, built block by block.

UNIT 1 • FINAL PROJECT

The number detective

★ FINAL PROJECT

The number detective

A mysterious five-digit number has been left on the board, with two of its digits smudged out: 4 3 □ 7 □. Your job is to recover it, using only the tools of Unit 1, and to present your reasoning like a detective's case file.

Your case

The number is known to satisfy all three clues below.

- 1 **Clue one**
The number is divisible by 9.
- 2 **Clue two**
The number is divisible by 11.
- 3 **Clue three**
The number is divisible by 4.

What to hand in

The recovered five-digit number, with a full justification for each digit.

A short note on which clue you used first and why that order was clever.

A check that your number really does pass all three tests.

A second number, of your own invention, with two smudged digits and three clues, for a classmate to solve.

How it will be judged

Criterion	Weight	--- ---	Correct recovered number	40%	Quality of justification	30%	Elegance of the solving order	15%	Originality of the invented puzzle	15%
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UNIT 1 • CHECK YOUR PROGRESS

Work through these questions without looking back. They cover the whole unit.

- 1 Work out:
a. $-11 + 7$ b. $4 - (-9)$ c. $(-8) \times (-6)$ d. $54 \div (-9)$
- 2 Find two integers whose sum is -3 and whose product is -40 .
- 3 Find the missing number: $\square \times (-7) = 42$.
- 4 Write down all the common factors of 28 and 42.
- 5 Write down the multiples of 9 between 40 and 90, and find the lowest common multiple of 9 and 12.
- 6 Find the highest common factor of 48 and 180, and use it to simplify $48/180$.
- 7 a. Which number less than 100 is both a perfect square and a perfect cube?
a. Explain your reasoning.
- 8 The number $6\square36$ is divisible by 9 and by 4. Find the missing digit.
- 9 Complete the pattern and describe the rule: $1^3 = 1^2$, $1^3 + 2^3 = 3^2$,
 $1^3 + 2^3 + 3^3 = 6^2$, $1^3 + 2^3 + 3^3 + 4^3 = \square^2$.

UNIT 1 • EVALUATION

Be honest. For each outcome, colour the circle that matches how you feel: green means confident, amber means nearly there, red means needs more work.

I CAN ...	GREEN	AMBER	RED
add and subtract with positive and negative integers	☐	☐	☐
multiply and divide with positive and negative integers	☐	☐	☐
find the lowest common multiple of two numbers	☐	☐	☐
find the highest common factor and simplify a fraction	☐	☐	☐
use tests for divisibility to spot factors quickly	☐	☐	☐
connect square and cube numbers to their roots	☐	☐	☐

UNIT 1 · WHAT CAN YOU DO?

✓ WHAT CAN YOU DO?

- | | |
|---|---|
| ☐ Add and subtract across zero on the number line. | ☐ Use the divisibility tests for 2, 3, 4, 5, 6, 8, 9, 10 and 11. |
| ☐ Apply the sign rules for products and quotients. | ☐ Evaluate square roots and cube roots of perfect powers. |
| ☐ Estimate an answer before calculating it exactly. | ☐ Trap a square root between two consecutive whole numbers. |
| ☐ Find an LCM by listing multiples. | |
| ☐ Find an HCF by listing factors, and simplify a fraction with it. | |

■ UNIT 1 ROUND-UP

integer · inverse · inverse operation · number line · product · quotient · multiple · common multiple · lowest common multiple · digit · factor · common factor · highest common factor · divisible · test for divisibility · square number · cube number · index · square root · cube root · consecutive

<!--CALCS All values verified by executing Python before authoring: 1.1: $-7+12=5$; $5+(-9)=-4$; $-4-(-8)=4$; $-3+(-6)=-9$; $2-9=-7$; $-8+5=-3$; $2-(-7)=9$... wait $2-(-7)=9$ yes I think of x , $x+(-7)=2 \rightarrow x=9$; $\text{sum}=-10$ $\text{diff}=2 \rightarrow -4,-6$; $-5+14-(-3)=12$; pyramid $\text{top}=1$; $198+347=545$ est 550; $396-187+58=267$ 1.2: $-6 \times 8=-48$; $(-63)/(-7)=9$; $9 \times (-5)=-45$; $56/(-8)=-7$; $7 \times (-4)=-28$; $(-9) \times (-5)=45$; $(-42)/6=-7$; $? \times (-6)=24 \rightarrow -4$; $(-36)/?=4 \rightarrow -9$; pairs product -12 listed; $8 \times (3-9)=8 \times (-6)=-48=24-72$; $(-19.7) \times 5.1 \approx -100$, exact -100.47 1.3: multiples 7,11,15; $\text{LCM}(6,10)=30$; $\text{LCM}(5,7)=35$; $\text{LCM}(8,14)=56$; $\text{LCM}(4,6,9)=36$; common multiples 4&6<50: 12,24,36,48; buses $\text{LCM}(18,24)=72$ min 1.4: factors 28,45,54; $\text{HCF}(18,27)=9$; $\text{HCF}(16,40)=8$; $\text{HCF}(36,54)=18$; $\text{HCF}(42,70)=14$; $\text{HCF}(30,75)=15$; $18/27=2/3$; $16/40=2/5$; $36/54=2/3$; boxes $\text{HCF}(36,54)=18$; $\text{HCF} \times \text{LCM} = \text{product}$ (12,18):216 1.5: 91818 /2,3,9,11 yes /5 no; $5124\%6=0$, $7335\%6=3$, $40206\%6=0$; $5?42\%9 \rightarrow 2$; 11 from 3,4,5,6: 44 (e.g. $3465?$ alt $\text{sum}(3+6)-(4+5)=0$ yes $3465/11=315$); 44... div by 2,4 not 5; div by 2&5 $\rightarrow 10$; coprime reasoning 1.6: $9^2=81$; $11^2=121$; $6^3=216$; $\sqrt{196}=14$; $\sqrt{81}+\sqrt{25}=9+5=14$; $\sqrt[3]{1000}=10$; $\sqrt{90}$ between 9,10; $\sqrt{289}=17$; $\sqrt{(81+144)}=\sqrt{225}=15$ vs $\sqrt{81}+\sqrt{144}=9+12=21$ (not equal); factors of 36: 9 (odd); cubes pattern $1,9,36,100=1^2,3^2,6^2,10^2$ triangular Check your progress: $-11+7=-4$; $4-(-9)=13$; $(-8) \times (-6)=48$; $54/(-9)=-6$; $\text{sum}-3$ product $-40 \rightarrow -8,5$; $? \times (-7)=42 \rightarrow -6$; common factors 28,42: 1,2,7,14; multiples of 9 40-90: 45,54,63,72,81,90; $\text{LCM}(9,12)=36$; $\text{HCF}(48,180)=12 \rightarrow 4/15$; square&cube <100: 1 and 64 ($8^2,4^3$); 6d36 div by 9 & 4 $\rightarrow d=3$ unique ($15+3=18$; $36\%4=0$); cubes pattern $1,9,36,100=1^2,3^2,6^2,10^2$ Corrections applied during authoring (caught by Python verification):

- Q8 originally 6d34 ($34\%4=2$, impossible) $\rightarrow$ changed to 6d36 with $d=3$
- P1.5 Q1: $91818\%11=1$ (not divisible) $\rightarrow$ removed 11 from the test list
- P1.5 Q3: 5d42 div by 9 needs $d=7$ ($5+7+4+2=18$), not $d=2$

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<!--IMAGEJOBS [{"name":"u1_opener","size":"portrait_4_3","prompt":"Flat geometric editorial vector illustration, architectural, a sweeping number line receding into mountain peaks, grid dots and axis ticks, cool blue tones. Strict palette: deep slate-ink #14202B, signal blue #2563EB, warm ivory #FBF7EF, ochre #D98E2B. Absolutely no text, no

letters, no numbers, no words, no watermark."}, {"name": "u1_number_line", "size": "landscape_4_3", "prompt": "Flat geometric editorial vector illustration, a horizontal number line as a road stretching left and right into infinity, arrowheads, zero marked as a milestone, minimalist. Strict palette: deep slate-ink #14202B, signal blue #2563EB, warm ivory #FBF7EF, ochre #D98E2B. Absolutely no text, no letters, no numbers, no words, no watermark."}, {"name": "u1_thermometer", "size": "portrait_4_3", "prompt": "Flat geometric editorial vector illustration, a stylised thermometer rising above and dipping below a central zero line, geometric ticks, cold blue lower half warm ochre upper half. Strict palette: deep slate-ink #14202B, signal blue #2563EB, warm ivory #FBF7EF, ochre #D98E2B. Absolutely no text, no letters, no numbers, no words, no watermark."}, {"name": "u1_factor_tree", "size": "landscape_4_3", "prompt": "Flat geometric editorial vector illustration, an abstract factor tree or number grid with branching geometric blocks, isometric, architectural. Strict palette: deep slate-ink #14202B, signal blue #2563EB, warm ivory #FBF7EF, ochre #D98E2B. Absolutely no text, no letters, no numbers, no words, no watermark."}, {"name": "u1_cubes", "size": "landscape_4_3", "prompt": "Flat geometric editorial vector illustration, stacked cubes forming a growing staircase of square and cube blocks, isometric, clean shadows. Strict palette: deep slate-ink #14202B, signal blue #2563EB, warm ivory #FBF7EF, ochre #D98E2B. Absolutely no text, no letters, no numbers, no words, no watermark."}] -->

GLOSSARY

Unit 1

The mathematical words this unit has taught you, gathered in one place.

common factor: a number that is a factor of two or more numbers.

common multiple: a number that is a multiple of two or more numbers.

consecutive: following one after another in order.

cube number: a number made by multiplying an integer by itself twice, such as $27 = 3^3$.

cube root: the number that cubes to give the value under the root, as in $\sqrt[3]{27} = 3$.

digit: any of the symbols 0 to 9 used to write numbers.

divisible: able to be divided exactly, with no remainder.

factor: a number that divides another exactly.

highest common factor (HCF): the largest common factor of two or more numbers.

index: the small raised number showing a power, as in 5^2 .

integer: a positive or negative whole number, or zero.

inverse: the number that undoes another; the inverse of 4 is -4 .

inverse operation: an operation that reverses another; subtraction is the inverse operation of addition.

lowest common multiple (LCM): the smallest common multiple of two or more numbers.

multiple: the result of multiplying a number by a positive integer.

number line: a line on which numbers are placed in order.

product: the result of a multiplication.

quotient: the result of a division.

square number: a number made by multiplying an integer by itself, such as $36 = 6^2$.

square root: the number that squares to give the value under the root, as in $\sqrt{36} = 6$.

test for divisibility: a quick rule that tells you whether one number divides another without doing the division.

SOURCES & REFERENCES

Where everything came from

Every fact, historical note and case study in this unit is real, and can be checked here. Links were live as of July 2026.

S UNIT 1 • INTEGERS

- 1 **Serra da Estrela.** The highest mountain range in mainland Portugal, used for the temperature scenario in Topic 1.1. Highest point Torre, 1,993 m; snow and freezing temperatures at altitude in winter. <https://en.wikipedia.org/wiki/Serra_da_Estrela>
- 2 **Casting out nines.** The historical arithmetic check mentioned in the "Did you know?" panel of Topic 1.5, twin of the divisibility rule for 9, described by Fibonacci in *Liber Abaci* (1202) and probably older, from the Indian subcontinent. <<https://mathworld.wolfram.com/CastingOutNines.html>>
- 3 **The history of zero and negative numbers.** Background to the opening essay of this unit. <https://en.wikipedia.org/wiki/Negative_number>
- 4 **Cascais.** The town used for the bus-headway LCM problem in Practise 1.3. The figures (18 and 24 minutes) are original to this book; the answer (72 minutes) is the verified lowest common multiple.
- 5 **Pastéis de nata and broas de mel.** Traditional Portuguese pastries used in the packing problem of Practise 1.4. The figures (36 and 54) are original; the answer (18) is the verified highest common factor.

✦ A NOTE ON HOW THIS BOOK WAS MADE

Every worked-example answer and every exercise value in this book was checked by a computer before it went to print. If you find an error, we want to know: write to the mathematics team via www.primeschool.pt.

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